Name: ____________________________ Date: ________________ Period: ________

Primary Objective

By the end of this lesson, I will be able to...

1. Construct a polynomial or piecewise function model using restrictions, transformations, or regression. *(LO 1.14.A)*
2. Construct a rational function model from an inverse-proportion context. *(LO 1.14.B)*
3. Apply a function model to predict values and compute average rates of change with appropriate units. *(LO 1.14.C)*

Vocabulary & Key Concepts

function model:

A function constructed to represent a real-world or mathematical scenario; used to make predictions.

domain restriction:

A constraint on the input values of a model based on the context (e.g., length must be positive).

inverse proportion:

A relationship where one quantity equals a constant divided by a power of another: $y = k/x^n$.

average rate of change:

$\frac{f(b) - f(a)}{b - a}$**; the slope of the secant line between $(a, f(a))$ and $(b, f(b))$.**

regression:

A technology-assisted process for finding the best-fit curve of a given type through a data set.

Hook

A farmer has exactly 80 feet of fencing and wants to build the biggest rectangular pen possible. One side of the pen sits against a barn, so no fencing is needed on that side. **Can a function tell us how to maximize area?**

Let w = width (in feet) of the pen. Draw a quick label: two sides of width w and one side of length ℓ, with the barn on the fourth side.

Question: What equation relates w and ℓ to the 80 feet of fencing?

$2w + \ell = 80$ ....

Section 1 — Building a Model from Restrictions (LO 1.14.A)

Setup. A rectangular pen has one side against a barn (no fencing needed there). Total fencing available: 80 ft. Two widths and one length must use all 80 ft.

Step 1. Write the fencing constraint: $2w + \ell = \mathbf{80}$

Step 2. Solve for ℓ in terms of w: $\ell = 80 - 2w$

Step 3. Area of the pen: $A = w \cdot \ell$. Substitute your expression for ℓ:

$$A(w) = w \cdot \big(\mathbf{80 - 2w}\big) = \mathbf{-2w^2 + 80w}$$

Step 4. Identify the **domain restriction**. Width must be positive and fencing constraint must hold: Domain: $w \in ($ **0** , **40** $)$

Step 5. What is the shape of $A(w)$? Circle one: **Linear** **Quadratic** **Cubic**
Because the leading coefficient is -2, the parabola opens **downward**.

Step 6. The maximum area occurs at the vertex. For $A(w) = aw^2 + bw$:

$$w_{\max} = -\frac{b}{2a} = -\frac{80}{2(-2)} = 20 \textbf{ ft} \implies A(w_{\max}) = -2(400) + 80(20) = 800 \text{ ft}^2$$

Teacher Note

$A(w) = -2w^2 + 80w$; vertex at $w = 20$ ft, $A(20) = 800$ ft^2. Domain: $0 < w < 40$.

Section 2 — Rational Function Model: Inverse Proportion (LO 1.14.B)

Context. Two electrically charged objects exert a force on each other. Experiments show the force is **inversely proportional to the square of the distance** between them.

This means: $F \propto \frac{1}{d^2}$, so we write $F = \frac{k}{d^2}$ for some constant $k > 0$.

Model: $r(d) = \frac{k}{d^2}$

Domain restriction: d = distance in meters, so $d > 0$.

End behavior (as inputs grow large or approach zero):

As $d \to 0^+$	$r(d) \to +\infty$ (force becomes very **large**)
As $d \to \infty$	$r(d) \to 0$ (force becomes very **small/negligible**)

Example. Suppose $k = 900$ (units: N·m^2). Complete the table:

d (m)	1	3	5	10
$r(d) = 900/d^2$	**900**	**100**	**36**	**9**

Section 3 — Applying the Model (LO 1.14.C)

Return to the pen model: $A(w) = -2w^2 + 80w$ on $(0, 40)$.

(a) Evaluate at specific inputs.
$A(10) = -2(\mathbf{10})^2 + 80(\mathbf{10}) = -200 + 800 =$ **600** ft^2

$A(20) = -2(\mathbf{20})^2 + 80(\mathbf{20}) = -800 + 1600 =$ **800** ft^2

(b) Average rate of change of area on $[10, 20]$.
ARC $= \dfrac{A(20) - A(10)}{20 - 10} = \dfrac{\mathbf{200}}{\mathbf{10}} = \mathbf{20}$
Units: **ft^2 per ft (square feet per foot of width)**

Interpret: On average, for each additional foot of width between $w = 10$ ft and $w = 20$ ft, the area **increases** by **20** square feet.

(c) Predict the width at maximum area.
From Section 1, the maximum occurs at $w =$ **20** ft with area $=$ **800** ft^2.
The length at that width is $\ell = 80 - 2(\mathbf{20}) =$ **40** ft.

Conclusion: The optimal pen is **20** ft wide and **40** ft long.